St. Catharines Chess Club Manager — User & Technical Guide

This document provides a comprehensive overview of how the Chess Club Weekly Manager application functions, including its user workflow, pairing logic, rating updates, King of the Hill leaderboard calculations, data persistence, and exact mechanics for fetching player ratings from external chess platform API endpoints.

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1. Overview & Architecture

The application is built as a standalone single-file web application (`index.html`). It contains HTML layout, CSS styling, and JavaScript logic in one file, allowing it to run entirely in any modern browser without needing a dedicated backend server or database setup.

- **Frontend Framework:** Plain Vanilla JavaScript with Tailwind CSS for modern responsive styling.
- **Storage Model:** Browser `localStorage` for automatic local data persistence.
- **External APIs:** Public, read-only REST endpoints provided by public chess statistics endpoints.

2. How the Chess.com Rating Integration Works

API Endpoint & Request Format

When a user clicks "Fetch Chess.com Rating" (or during player creation with a username specified), the application sends a standard HTTP GET request directly to the public REST API endpoint for stats.

- **User Input Sanitization:** The application strips extra whitespace and converts the username to lowercase (e.g., `" Hikaru "` → `"hikaru"`).
- **Authentication:** No API key or secret token is required. The endpoint is completely open and free to query.

Data Extraction Logic

Upon a successful JSON response (HTTP 200), the application extracts ratings based on the selected preferred time control:

1. **Rapid Rating Preference (Default):** Checks rapid rating fields.
2. **Blitz Fallback:** Checks blitz rating fields.
3. **Bullet Fallback:** Checks bullet rating fields.
4. **Manual Fallback:** If the account exists but has no active rating in those categories (or if the user profile isn't found), the application alerts the user and defaults to a configurable starting Elo (e.g., 1200).

3. Player Management & Registration

- **Adding Players:** Club organizers can add members manually by entering a full name and an initial Elo rating, or by entering a Chess.com username to fetch their rating automatically.
- **Attendance Toggle:** Each player in the registry features a "Present" checkbox. Only active/present players are included when generating pairings for the current week's round.
- **Player Deletion:** Removing a player purges them from the active list while retaining historical match data in previously saved rounds.

4. Weekly Attendance & Matchmaking (Pairing Algorithm)

When the organizer clicks "Generate Round Pairings", the application executes a proximity-based Swiss-style pairing algorithm:

1. **Filtering:** Selects all players currently checked as Present.
2. **Bye Handling:** If an odd number of players are present, the lowest-rated player who has not yet received a bye in previous rounds is awarded a Bye (+0.5 points / no Elo shift) for that round.
3. **Elo Sorting:** The remaining active players are sorted in descending order by their current club Elo rating.
4. **Proximity Matching with Repeat Prevention:**
    - The top-rated unassigned player is matched with the nearest available opponent in the sorted list.
    - **Rematch Prevention:** The engine checks historical rounds to see if the two players have already faced each other. If they have, it evaluates the next closest opponent to avoid repetitive weekly matchups.
5. **Manual Overrides:** Organizers can manually alter color assignments (White vs. Black) or adjust match pairs before submitting round results.

5. Round Management & Game Result Processing

Each active pairing presents three outcome options:

- **White Win** (1 - 0)
- **Draw** (0.5 - 0.5)
- **Black Win** (0 - 1)

Once all active games have a recorded outcome, the organizer clicks "Finish & Close Round". Closing the round triggers the dynamic rating engine and records the results into the permanent club history log.

6. Dynamic Elo Calculation Engine

When a round is closed, player ratings are updated using the standard FIDE Elo Rating System with a default K-factor of 32:

1. Expected Score Formula

For two players with ratings R_a and R_b, the expected score E_a for Player A is:

$$E_a = \frac{1}{1 + 10^{(R_b - R_a) / 400}}$$

2. Rating Update Formula

After the game result S_a is recorded (1.0 for a win, 0.5 for a draw, 0.0 for a loss), the new rating R'_a is calculated as:

$$R'_a = R_a + K \times (S_a - E_a)$$

- **K-Factor:** Fixed at 32 for fast, dynamic rating adjustments suited for weekly club play.
- **Simultaneous Updates:** Both players' ratings update symmetrically based on their pre-game ratings.

7. King of the Hill Leaderboard Mechanics

The King of the Hill Leaderboard dynamically ranks club members based on performance metrics:

1. **Summit / Current King:** The highest-rated active player holds the top spot on the leaderboard hill.
2. **Win-Loss-Draw Tally:** Tracks total games played, total victories, defeats, and draws across all recorded weekly rounds.
3. **Streak Tracking:** Tracks active winning or undefeated streaks for players.
4. **Sort Priority:**
    - **Primary:** Current Club Elo Rating.
    - **Secondary:** Win Percentage (Wins / Total Games).
    - **Tertiary:** Head-to-head history or total wins.

8. Data Storage, Import & Export

Local Storage (`localStorage`)

All club data—player rosters, current Elo ratings, attendance history, and past round pairings—is saved automatically in the browser's `localStorage` under a dedicated key (chess_club_manager_data). This ensures data persists across page refreshes and browser restarts without needing a server.

Backup & Restore (JSON Export/Import)

- Export: Generates and downloads a .json file containing the complete state of the club database.
- Import: Allows uploading a saved .json file to restore player lists and history onto another computer or browser.